European Commission

Defence Industry and Space

Drones and Counter-Drone Systems

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DEFENCE APPLICATIONS

Drones are redefining the modern battlefield, enabling real-time intelligence, precision strikes, and unprecedented situational awareness

Mastering the interplay between drones and counter-drone systems is key for Europe to safeguard its borders and preserve its technological edge

Counter-drone systems have become indispensable to detect, disrupt, and neutralise hostile threats across all domains

Why do drones and counter-drone systems matter?

Drones, or Unmanned Aerial Vehicles (UAVs), have become indispensable on the modern battlefield, providing real-time intelligence, enabling precision strikes, and contributing directly to the neutralisation or destruction of enemy personnel, materiel and installations. They **operate at all levels of engagement**, from small quadcopters used by infantry units to larger systems conducting strategic surveillance or long-range strikes. This has made drones central to situational awareness and command-and-control.

At the same time, their proliferation has created new vulnerabilities. **Low-cost drones can threaten and destroy high-value classic military assets** and civilian infrastructure. Because they are cheap and easily replaced, they give smaller actors disproportionate power. Counter-drone systems, or C-UAS, play an essential defensive role: they protect airspace, troops, equipment, and critical sites from an increasingly diverse range of threats.

Key differences between civil and defence drone technologies

The distinction between civil and defence drones is not merely technical but also operational. Civil drones perform routine tasks, whereas **military drones are designed for contested, high-risk environments** where survivability, redundancy and adaptability are critical for mission success.

Civil drones, used for mapping, inspection or delivery, operate in controlled environments under predictable conditions. Defence drones, by contrast, must **function in hostile settings** where communications can be jammed, navigation systems spoofed, and air defences active. They require hardened communication links, autonomous navigation capabilities, and resilience to electronic warfare and cyber interference.

Military drones are also **integrated into broader defence networks**, providing real-time data to intelligence, surveillance and targeting systems. Their software supports secure communications and rapid operational decision-making, rather than remote control.

Main advantages of using drones and C-UAS

- Enhanced reconnaissance and surveillance
- Rapid production and deployment
- Real-time situational awareness
- Reduced risk to human operators
- Precision targeting and strike capability
- Cost-effective force multiplier

Main vulnerabilities arising from the use of drones and C-UAS

- Vulnerability to electronic warfare and hacking
- Risk of collateral damage or misidentification
- Dependence on GPS and communication links
- Escalation of autonomous warfare
- Proliferation to non-state actors
- Ethical and legal accountability gaps

When will EDF project results be translated into military capabilities?

The **EU Defence Readiness Roadmap 2030** identifies drones and counter-drone systems as one of the key capability areas for strengthening Europe's strategic autonomy and operational readiness. In this context, the **European Drone Defence Initiative** and the **Eastern Flank Watch** have been launched as flagship projects to reinforce air and border defence, protect critical infrastructure, and enhance resilience against unmanned and hybrid threats.

Complementing these efforts, key EDF actions are strengthening EU leadership in this domain by **building a connected ecosystem** that links research, development, and would facilitate scalable production. A total of **€1 billion** from the EDF and its precursor programmes has been dedicated to a wide range of drone-related research and development actions.

Successful project examples include **LOTUS**, **ALTISS**, **HYBRID**, **REACT II**, **E-CUAS** and **SMALL UAS**. The HYBRID project focuses on developing a long-endurance, **hydrogen fuel cell-powered electric drone**. REACT II builds on the results of REACT I to advance Airborne Electronic Attack (AEA) capabilities, equipping the EU with critical **electronic warfare systems** to counter drone threats. E-CUAS, the successor to the successful JEY-CUAS initiative, is developing **next-generation counter-drone capabilities** by integrating multiple technologies, including passive and active sensors, soft- and hard-kill effectors, command and control components, networking and communications infrastructure, cybersecurity features, and interoperability data models. Finally, the SMALL UAS project is developing an advanced, **AI-driven swarm system** composed of unmanned aerial systems designed for multi-domain operations.

Projects from precursor programme EDIDP delivering first results

MALE RPAS
European Medium Altitude Long Endurance Remotely Piloted Aircraft System
EU financing: €98.115M

LOTUS
Low Observable Tactical Unmanned air System
EU financing: €8.7M

ALTISS
Autonomous Light Intelligence & Surveillance System = Highly Automated Swarm of Affordable ISR Long Endurance UAVs
EU financing: €2.5M

EDF projects expected to deliver results by 2030

HYBRID
Hydrogen Battlefield Reconnaissance and Intelligence Drone
EU financing: €3.1M

REACT II
Responsive Electronic Attack for Cooperation Tasks II
EU financing: €40M

Small UAS
Small UAS
EU financing: €10.99M

E-CUAS
European Counter Unmanned Aerial Systems
EU financing: €43M



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